

KARLENS NIKITA PIERRE

knpbiz9@gmail.com | (908) 265-6280 | [linkedin.com/in/karlenspierre](https://www.linkedin.com/in/karlenspierre) | github.com/Kaarsz | karlensnpierre.netlify.app

PROFILE

Computer Science student (Cybersecurity minor) with hands-on experience in backend development, cloud infrastructure, applied AI/LLM systems, and security research, seeking Software Engineering and Cybersecurity internships.

EDUCATION

Villanova University

B.S. in Computer Science, Minor in Cybersecurity | GPA: 3.2

Villanova, PA

Expected May 2027

Relevant Coursework: Operating Systems, Principles of Database Systems, Data Structures & Algorithms, Statistics for Computing

PROFESSIONAL EXPERIENCE

Stealth Startup

Software Engineering Intern

Jersey City, NJ

June 2026 – August 2026

- Contributed to an existing production digital infrastructure, improving system latency by 19% through targeted performance and reliability enhancements.
- Leveraged an internal, proprietary AI agent built for the company's engineering workflows to accelerate development and streamline technical tasks.
- Gained hands-on experience with large language models optimized through retrieval-augmented generation (RAG) and fine-tuning, deepening applied understanding of modern LLM architectures.

Amazon

Software Development Engineer Fellow

Jersey City, NJ

June 2025 – August 2025

- Engineered and tested a backend order-processing service using Git-based version control, automated unit tests, and systematic debugging workflows to improve reliability and reduce defects.
- Built a Java REST API for inventory management, applying object-oriented design and efficient data structures to increase daily request-handling capacity by 33%.
- Architected an AWS-based file processing pipeline grounded in distributed-systems principles, improving performance, fault tolerance, and scalability.

TECHNICAL PROJECTS & RESEARCH

Endpoint Security Research Tool

Independent Research — Isolated Lab Environment

Villanova, PA

April 2024

- Built a Python-based research tool simulating insider-threat and data-exfiltration techniques (keystroke, clipboard, and system-metadata capture) in a controlled, isolated environment to study attacker tradecraft.
- Implemented automated data-transfer workflows using Python's email libraries and SMTP to model exfiltration communication channels.
- Integrated microphone and webcam capture using ImageGrab and sounddevice to evaluate endpoint-monitoring detection gaps.
- Developed HTTP communication modules with Python's requests library to test remote data-transmission and system-interaction scenarios.

Cyber Warfare Research Paper

Author

West Orange, NJ

October 2022

- Conducted analytical research on modern cyber-warfare threats, ransomware ecosystems, and malware architectures, including Trojans, botnets, worms, and rootkits.
- Analyzed historical cyberattacks to identify common exploitation patterns and produced a technical report outlining defensive strategies and mitigation techniques.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript/TypeScript, SQL, HTML/CSS, Assembly

Frameworks/Tools: React, Git, Linux, AWS, PyTorch

Concepts: OOP, CI/CD, Unit Testing, REST APIs, Data Structures & Algorithms, Debugging

AI/ML: LLMs, Retrieval-Augmented Generation (RAG), Fine-Tuning

LEADERSHIP & EXTRACURRICULAR

Alpha Phi Alpha Fraternity, Inc.

Vice President, Treasurer & Financial Secretary

Philadelphia, PA

May 2024 – Present

- Co-led chapter operations for the Mighty Psi Chapter, recognized as Chapter of the Year among 17 chapters in the Eastern Region.
- Spearheaded fundraising efforts that generated over \$10,000 for the chapter treasury.
- Organized 33 events and delegated responsibilities across seven universities in the charter.

Black Student Union

Vice President

Philadelphia, PA

May 2024 – May 2026

- Serve as a liaison between students, faculty, and administration at Villanova University.
- Advocate for the interests and viewpoints of approximately 400 undergraduate students.